



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - 3

Time : 3 Hrs. 20 Mins.

Answers

1. (1)	41. (2)	81. (3)	121. (1)	161. (2)
2. (2)	42. (2)	82. (1)	122. (4)	162. (4)
3. (4)	43. (3)	83. (1)	123. (2)	163. (2)
4. (1)	44. (2)	84. (2)	124. (3)	164. (3)
5. (4)	45. (1)	85. (3)	125. (4)	165. (1)
6. (3)	46. (2)	86. (3)	126. (2)	166. (4)
7. (4)	47. (4)	87. (2)	127. (2)	167. (1)
8. (3)	48. (2)	88. (1)	128. (2)	168. (3)
9. (3)	49. (3)	89. (3)	129. (3)	169. (3)
10. (1)	50. (3)	90. (2)	130. (4)	170. (4)
11. (2)	51. (3)	91. (2)	131. (2)	171. (4)
12. (4)	52. (2)	92. (3)	132. (1)	172. (2)
13. (4)	53. (1)	93. (3)	133. (4)	173. (2)
14. (4)	54. (3)	94. (1)	134. (2)	174. (2)
15. (2)	55. (1)	95. (3)	135. (1)	175. (3)
16. (3)	56. (1)	96. (1)	136. (4)	176. (2)
17. (1)	57. (1)	97. (2)	137. (4)	177. (1)
18. (3)	58. (3)	98. (2)	138. (2)	178. (2)
19. (3)	59. (3)	99. (4)	139. (2)	179. (4)
20. (1)	60. (1)	100. (1)	140. (3)	180. (1)
21. (1)	61. (3)	101. (4)	141. (3)	181. (3)
22. (3)	62. (2)	102. (2)	142. (1)	182. (3)
23. (1)	63. (1)	103. (3)	143. (3)	183. (4)
24. (3)	64. (4)	104. (2)	144. (3)	184. (1)
25. (3)	65. (2)	105. (3)	145. (3)	185. (4)
26. (1)	66. (1)	106. (3)	146. (2)	186. (3)
27. (2)	67. (2)	107. (2)	147. (2)	187. (1)
28. (2)	68. (3)	108. (2)	148. (2)	188. (3)
29. (3)	69. (2)	109. (1)	149. (4)	189. (4)
30. (2)	70. (3)	110. (4)	150. (1)	190. (2)
31. (2)	71. (1)	111. (3)	151. (3)	191. (1)
32. (4)	72. (2)	112. (2)	152. (4)	192. (2)
33. (4)	73. (2)	113. (3)	153. (2)	193. (4)
34. (2)	74. (4)	114. (3)	154. (1)	194. (4)
35. (1)	75. (1)	115. (1)	155. (4)	195. (3)
36. (2)	76. (1)	116. (3)	156. (3)	196. (3)
37. (1)	77. (1)	117. (1)	157. (2)	197. (4)
38. (1)	78. (4)	118. (1)	158. (4)	198. (3)
39. (1)	79. (3)	119. (2)	159. (4)	199. (2)
40. (1)	80. (3)	120. (4)	160. (1)	200. (4)



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Answers and Solutions

PHYSICS

SECTION - A

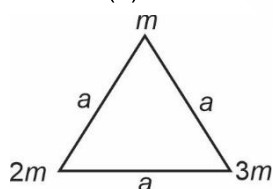
1. Answer (1)

$$g = \frac{Gm}{r^2}$$

$$\frac{g}{G} = \frac{m}{r^2}$$

Hence, unit of $\frac{g}{G}$ is $\frac{\text{kg}}{\text{m}^2}$

2. Answer (2)



$$U = \frac{-2Gm^2}{a} - \frac{6Gm^2}{a} - \frac{3Gm^2}{a}$$

$$= \frac{-11Gm^2}{a}$$

3. Answer (4)

$$T.E = -E$$

$$B.E = E$$

4. Answer (1)

As per Kepler's law of time period

$$T^2 \propto R^3$$

5. Answer (4)

$$V \times \rho_b g = V_{in} \times \rho_\ell g$$

$$\frac{\rho_b}{\rho_w} = \left(\frac{V_{in}}{V} \right)$$

$$= \frac{70}{100}$$

$$= 0.7$$

6. Answer (3)

$$P_B = P_A + \rho g_{eff} h$$

$$P_B = P_A + \rho \frac{4gh}{3}$$

$$P_B - P_A = \left(\frac{4\rho gh}{3} \right)$$

7. Answer (4)

Water droplets are spherical in free fall condition due to surface tension.

8. Answer (3)

$$\Delta \ell = \frac{F\ell}{YA}$$

$$\Delta \ell \propto \ell \quad (F, Y \text{ and } A \text{ are same})$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = \frac{\ell_1}{\ell_2} = \frac{2}{3}$$

9. Answer (3)

Longitudinal strain is meaningful in solid.

10. Answer (1)

Steel is more elastic than rubber and hooke's law is applicable up to elastic limit.

11. Answer (2)

$$\frac{K-273}{100} = \frac{F-32}{180}$$

$$K-273 = \frac{5}{9}(F-32)$$

$$K_1 = \frac{5}{9}(F_1-32) + 273$$

$$K_2 = \frac{5}{9}(F_2-32) + 273$$

$$K_2 - K_1 = \frac{5}{9}(F_2 - F_1)$$

$$50 = \frac{5}{9}n$$

$$n = 90$$

12. Answer (4)

$$\frac{130-50}{R} + \frac{20-50}{R} + \frac{x-50}{R} = 0$$

$$x = 0^\circ\text{C}$$

13. Answer (4)

$$\begin{aligned}\Delta P &= \frac{2T}{R} \\ &= \frac{2 \times 0.072}{1.2 \times 10^{-3}} \\ &= \frac{2 \times 7.2 \times 10^{-2}}{1.2 \times 10^{-3}} \\ &= 2 \times 6 \times 10 \\ &= 120 \text{ N/m}^2\end{aligned}$$

14. Answer (4)

$$h = \frac{2s \cos \theta}{\rho g r}$$

$$\frac{h_1}{h_2} = \frac{r_2}{r_1} = 2 : 1$$

15. Answer (2)

The working of venturimeter is based on Bernoulli's theorem.

16. Answer (3)

$$R = 2\sqrt{h(H-h)}$$

$$\text{For } R = R_{\max}$$

$$\frac{dR}{dh} = 0$$

$$h = \frac{H}{2}$$

17. Answer (1)

From mechanical energy conservation

$$\frac{1}{2}mv^2 - \frac{3GMm}{2R} = 0 \Rightarrow v = \sqrt{\frac{3GM}{R}} = \sqrt{1.5} v_e$$

18. Answer (3)

$$F_1 = F_2 = \frac{G(1)(1)}{(2)^2} = \frac{G}{4}$$

$$\vec{F}_{\text{net}} = F_1(\hat{i}) + F_2(\hat{j}) = F(\hat{i} + \hat{j}) = \frac{G}{4}(\hat{i} + \hat{j}) \text{ N}$$

19. Answer (3)

As the height decreases, the rate of flow with which the water is coming out decreases.

20. Answer (1)

As the cork moves up, the force due to buoyancy remains constant. As its speed increases, the retarding force due to viscosity increases, being proportional to the speed. Thus the acceleration gradually decreases.

21. Answer (1)

$$\text{Velocity of efflux} = \sqrt{2gh} = \sqrt{2 \times 10 \times 5} = 10 \text{ m/s}$$

$$\text{Rate of flow} = Av = (1 \times 10^{-4}) \times 10 = 10^{-3} \text{ m}^3/\text{s}$$

22. Answer (3)

$$r_{\text{new}} = \sqrt{r_1^2 + r_2^2} = \sqrt{3^2 + 4^2} = 5 \text{ cm}$$

23. Answer (1)

$$t = \frac{(P_t - P_0)}{(P_{100} - P_0)} \times 100^\circ\text{C} = \frac{(60 - 50)}{(90 - 50)} \times 100 = 25^\circ\text{C}$$

24. Answer (3)

When the temperature of a liquid is increased by $\Delta T^\circ\text{C}$ the mass will remain unchanged while due to thermal expansion volume will increase and become V' .

$$V' = V(1 + \gamma\Delta T)$$

where γ is the coefficient of volume expansion of liquid.

$$\rho' = \frac{m}{V'} = \frac{m}{V(1 + \gamma\Delta T)} = \frac{\rho}{1 + \gamma\Delta T} \quad \left[\because \rho = \frac{m}{V} \right]$$

$$\text{Fractional change in density} = \frac{\rho - \rho'}{\rho} = \frac{\Delta\rho}{\rho} = \frac{\delta v}{v}$$

$$= \gamma\Delta T$$

$$= 49 \times 10^{-5} \times 30$$

$$= 1.5 \times 10^{-2}$$

25. Answer (3)

The densest layer of water will be at bottom. The density of water is maximum at 4°C . So the temperature of bottom of lake will be 4°C .

26. Answer (1)

$$R = R_1 + R_2$$

$$\frac{2\ell}{k_{\text{eff}}A} = \frac{\ell}{kA} + \frac{\ell}{3kA}$$

$$\frac{2\ell}{k_{\text{eff}}} = \frac{\ell}{k} + \frac{\ell}{3k}$$

$$\frac{2}{k_{\text{eff}}} = \frac{4}{3k}$$

$$K_{\text{eff}} = \frac{3k}{2}$$

27. Answer (2)

$$\Delta\ell = \frac{FL}{AY} \Rightarrow \Delta\ell \propto \frac{L}{r^2}$$

28. Answer (2)

By Bernoulli's theorem, $\frac{1}{2}\rho v_1^2 + P_1 = \frac{1}{2}\rho v_2^2 + P_2$

$$P_1 - P_2 = \frac{1}{2}\rho v_2^2 - \frac{1}{2}\rho v_1^2$$

$$2h\rho g = (v_2^2 - v_1^2)\rho$$

$$h = \frac{v_2^2 - v_1^2}{2g} = 0.8 \text{ m}$$

29. Answer (3)

$$(l_1 - l_0) = \frac{T_1 l_0}{AY} \quad \dots \text{eq. (1)}$$

$$(l_2 - l_0) = \frac{T_2 l_0}{AY} \quad \dots \text{eq. (2)}$$

Equation (1) divided by (2)

$$\frac{l_1 - l_0}{l_2 - l_0} = \frac{T_1}{T_2}$$

$$T_2 l_1 - T_2 l_0 = T_1 l_2 - T_1 l_0$$

$$T_2 l_0 - T_1 l_0 = T_2 l_1 - T_1 l_2$$

$$l_0 (T_2 - T_1) = T_2 l_1 - T_1 l_2$$

$$l_0 = \left(\frac{T_2 l_1 - T_1 l_2}{T_2 - T_1} \right)$$

30. Answer (2)

The Young's modulus of the material of a wire is equal to the stress required to produce unit strain.

31. Answer (2)

$$g' = \frac{GM}{r^2} = \frac{GM}{(R+h)^2}$$

$$\text{Now, } g' = \frac{g}{9}$$

$$\Rightarrow \frac{GM}{(R+h)^2} = \frac{1}{9} \frac{GM}{R^2}$$

$$\Rightarrow h = 2R$$

32. Answer (4)

$$T^2 \propto r^3$$

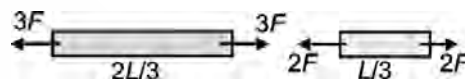
$$\left(\frac{84}{T} \right)^2 = \left(\frac{r}{3r} \right)^3$$

$$\Rightarrow T = 3\sqrt{3} \times 84$$

$$= 435.96$$

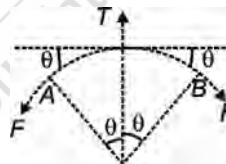
$$\approx 436 \text{ days}$$

33. Answer (4)



$$I = \frac{3F \left(\frac{2L}{3} \right)}{AY} + \frac{2F \left(\frac{L}{3} \right)}{AY} \Rightarrow I = \frac{8FL}{3AY}$$

34. Answer (2)



F = Tension in the thread

$$2F \sin\theta = 2T(AB) \quad (\because \sin\theta = \theta)$$

$$2F\theta = 2T(2R\theta)$$

$$F = 2RT$$

35. Answer (1)

$$v_e = \sqrt{2gR}, \text{ Velocity at infinity is zero.}$$

SECTION-B

36. Answer (2)

According to Kepler's law of areal velocity

$$\frac{dA}{dt} = \frac{L}{2m}$$

37. Answer (1)

At any point below the surface of earth

$$g = \frac{GM}{R^3} r$$

$$g \propto r$$

38. Answer (1)

Potential energy at the surface of the earth

$$= -\frac{GMm}{R}$$

$$= -gmR$$

Potential energy at a distance R from the surface

$$\text{of earth} = -\frac{GMm}{R+R}$$

$$= -\frac{GMm}{2R}$$

$$= -\frac{mgR}{2}$$

$$\text{Gain in potential energy} = -\frac{gmR}{2} - (-gmR)$$

$$= \frac{gmR}{2}$$

39. Answer (1)

Since the escape velocity is independent of direction of projection, therefore escape velocity of body, projected at an angle 45° with the vertical will be same i.e. 11 km/s.

40. Answer (1)

$$\frac{\Delta \ell}{\ell} = \left(\frac{1}{100} \right)$$

$$V = \ell^3$$

$$\frac{\Delta V}{V} = 3 \frac{\Delta \ell}{\ell}$$

$$= 3 \times (0.01)$$

$$= 0.03$$

41. Answer (2)

$$a + r + t = 1$$

$$0.2 + 0.4 + t = 1$$

$$t = 0.4$$

$$Q = 400 \times 0.4 \times 10$$

$$= 1600 \text{ J}$$

42. Answer (2)

$$m_1 S_1 \Delta T = m_2 S_2 \Delta T$$

$$m_1 \times S_1 \times 4 = m_2 \times S_2 \times 6$$

$$\frac{S_1}{S_2} = \frac{m_2}{m_1} \times \frac{3}{2}$$

$$= \frac{1}{2} \times \frac{3}{2}$$

$$= \frac{3}{4}$$

43. Answer (3)

Terminal velocity of a spherical body in viscous fluid is

$$V_T = \frac{2}{9\eta} r^2 (\rho_s - \rho_l)$$

$$V_T \propto r^2$$

44. Answer (2)

Height of liquid column

$$h = \frac{2T \cos \theta}{r \rho g}$$

$$\text{If } h = 0$$

$$\therefore \theta = 90^\circ$$

45. Answer (1)

$$1000r^3 = R^3$$

$$r = \frac{R}{10}$$

$$A_1 = 4\pi r^2$$

$$A_2 = 1000 \times 4\pi r^2 = 40\pi R^2$$

$$\text{Change in energy} = (\Delta A)T$$

$$= 36\pi R^2 T$$

$$= 36\pi \times 10^{-4} \times 0.075$$

$$= 27\pi \times 10^{-5} \text{ J}$$

46. Answer (2)

$$\frac{v^2}{2g} = 10 \text{ cm}$$

$$\frac{v^2}{2 \times 10} = 10 \times 10^{-2} \text{ m}$$

$$v = \sqrt{2 \times 10^2 \times 10^{-2}}$$

$$v = \sqrt{2} \text{ m/s}$$

$$\simeq 1.4 \text{ m/s}$$

47. Answer (4)

$$\tan 37^\circ = Y_B$$

$$\tan 53^\circ = Y_A$$

$$\frac{Y_A}{Y_B} = \frac{\frac{4}{3}}{\frac{3}{4}} = \frac{16}{9}$$

$$Y_B = \frac{9}{16} Y_A$$

48. Answer (2)
The strip will bend having concave surface on side 1.

49. Answer (3)
Water will expand on heating as well as on cooling from 4°C.

50. Answer (3)

$$\omega^2 = \frac{GM}{r^3}$$

$$(\omega')^2 = \frac{GM}{(4r)^3}$$

$$\frac{\omega^2}{(\omega')^2} = \left(\frac{4r}{r}\right)^3 = 64$$

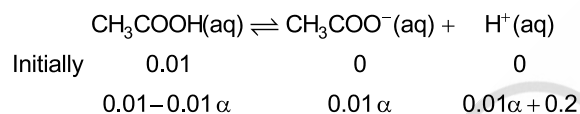
$$\frac{\omega}{\omega'} = 8$$

CHEMISTRY

SECTION - A

51. Answer (3)
Salt of strong acid and weak base $[(\text{NH}_4)_2\text{SO}_4]$ on hydrolysis form acidic solution.

52. Answer (2)



$$K_a = \frac{0.01\alpha(0.01\alpha + 0.2)}{0.01(1 - \alpha)} \alpha \ll 1$$

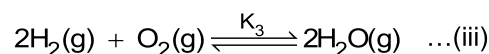
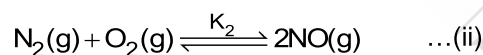
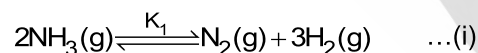
$$K_a = 0.2\alpha$$

$$\alpha = \frac{1.8 \times 10^{-5}}{0.2}$$

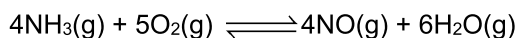
$$= 9 \times 10^{-5}$$

$$\therefore \% \text{ ionization} = 0.009\%$$

53. Answer (1)

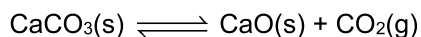


Applying $2 \times (\text{i}) + 2 \times (\text{ii}) + 3 \times (\text{iii})$, we get



$$K = K_1^2 K_2^2 K_3^3$$

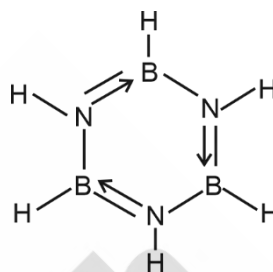
54. Answer (3)



$$\Delta n_g = (1) - (0) = 1$$

$$\therefore K_P = K_C(\text{RT})^{\Delta n_g} = K_C \text{RT}$$

55. Answer (1)



Number of sigma bonds = 12

Number of pi bonds = 3

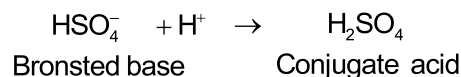
56. Answer (1)

On increasing temperature, dissociation of water increases means K_w increases but pH and pOH both decreases.

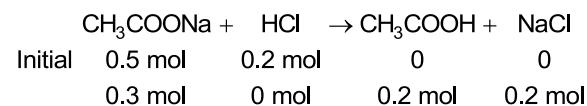
57. Answer (1)

$$K_C = \frac{[\text{D}_2]^3}{[\text{A}_2][\text{B}_2]^2}$$

58. Answer (3)



59. Answer (3)

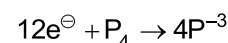


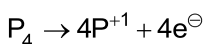
$$\text{Now, } \text{pH} = \text{p}K_a + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$= 4.74 + \log \frac{0.3}{0.2}$$

$$= 4.74 + \log \frac{3}{2} = 4.92$$

60. Answer (1)





$\therefore$ 4 mole $P_4 \Rightarrow$ 12 mole $e^{\ominus}$ gain or loss

$\therefore$ 1 mole $P_4 \Rightarrow \frac{12}{4} = 3$ mole $e^{\ominus}$

$$\text{Equivalent weight} = \frac{M}{3}$$

61. Answer (3)

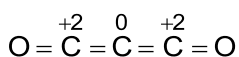
In H_2S , S is present in its lowest oxidation state *i.e.*, -2, so it cannot be further reduced.

62. Answer (2)

Balance reaction is



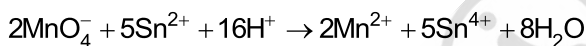
63. Answer (1)



64. Answer (4)

In $HClO_4$, Cl is at its maximum oxidation state *i.e.*, +7.

65. Answer (2)



66. Answer (1)

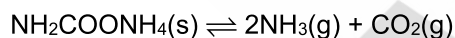
MnO_4^- can act as a self-indicator.

67. Answer (2)

Reduction potential $K < Zn < Cr < Ag$.

Therefore, Zn can reduce Cr^{3+} and Ag^+ but not K^+ .

68. Answer (3)

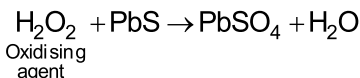
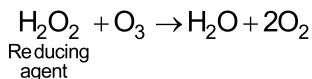


(i) $2P$ P
(f) P' $(2P)$

$$\therefore K_P = (P')^2(2P) = (2P)^2(P)$$

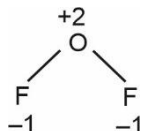
$$\Rightarrow P' = \sqrt{2}P = \frac{\sqrt{2}}{3} \times 3P = \frac{\sqrt{2}}{3} \times (P_{\text{total}})$$

69. Answer (2)



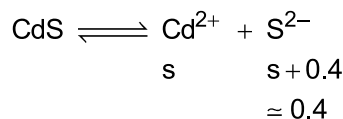
70. Answer (3)

In OF_2 , the oxygen is assigned an oxidation number of +2.



71. Answer (1)

Let the solubility of CdS be s mol/L



$$K_{sp} = [Cd^{2+}][S^{2-}]$$

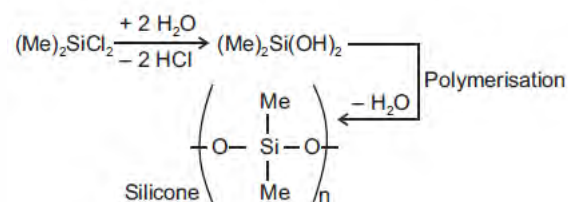
$$8 \times 10^{-27} = s(0.4)$$

$$s = 2 \times 10^{-26}$$

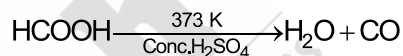
72. Answer (2)

Addition of $N_2(g)$ at equilibrium will shift the reaction in forward direction.

73. Answer (2)



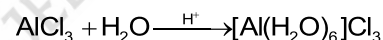
74. Answer (4)



75. Answer (1)

In H_3PO_4 , it is more difficult to remove a proton from a doubly charged HPO_4^{2-} anion as compared to $H_2PO_4^-$.

76. Answer (1)



Complex A is $[Al(H_2O)_6]^{3+}$ and B is sp^3d^2

77. Answer (1)

CO : Neutral

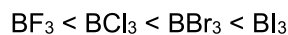
In_2O_3 : Basic

Ga_2O_3 : Amphoteric

SiO_2 : Acidic

78. Answer (4)

Order of Lewis acid nature:



79. Answer (3)

Graphite has layered structure with sp^2 hybridised carbon.

80. Answer (3)

$CaOCl_2$ exists as Ca^{2+} , Cl^- and OCl^- .

$\therefore$ O.S of chlorine in $CaOCl_2$ is -1 and +1.

81. Answer (3)

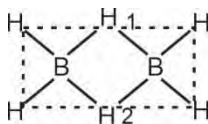
Solution of salt of weak acid and weak base acts as a buffer.

82. Answer (1)

At equilibrium, $\Delta G^\circ = -RT \ln K$

$$\begin{aligned}\Delta G^\circ &= -2.303 \times 2 \times 300 \times \log 10^3 \\ &= -2.303 \times 2 \times 300 \times 3 \log 10 \\ &= -2.303 \times 1800 \\ &= -4.14 \text{ kcal mol}^{-1}\end{aligned}$$

83. Answer (1)



Contain 2 B-H-B bonds.

84. Answer (2)

Pb^{2+} is more stable than Pb^{4+} (inert pair effect).

85. Answer (3)

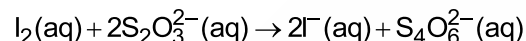
When air is used instead of steam, a mixture of CO and N_2 is produced, which is called producer gas.

SECTION-B

86. Answer (3)

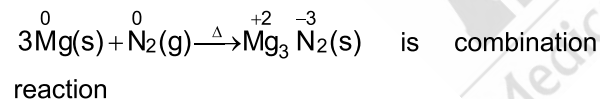
On adding CO(g) , $\text{CoCl}_2\text{(g)}$ will increase and $\text{PCl}_5\text{(g)}$ will decrease.

87. Answer (2)



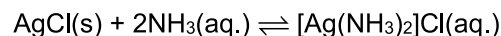
Iodine itself gives an intense blue colour with starch and has a very specific reaction with $\text{S}_2\text{O}_3^{2-}$.

88. Answer (1)



89. Answer (3)

Due to complex formation solubility of AgCl will be maximum in 0.1 M NH_3 and minimum in 0.5 M NaCl.



90. Answer (2)

$$\text{p}K_a = -\log K_a$$

For conjugate acid-base pair

$$K_a \times K_b = K_w$$

$$K_b = \frac{K_w}{K_a}$$

$$= \frac{10^{-14}}{1 \times 10^{-4}}$$

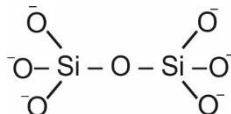
$$= 1 \times 10^{-10}$$

91. Answer (2)

CCl_4 cannot get hydrolysed due to absence of empty d -orbital in outermost shell of carbon atom.

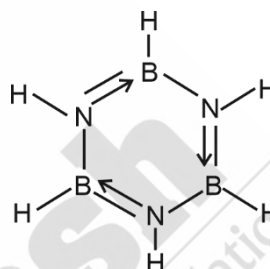
92. Answer (3)

Pyrosilicates have $\text{Si}_2\text{O}_7^{6-}$ unit



93. Answer (3)

Borazine is also known as inorganic benzene



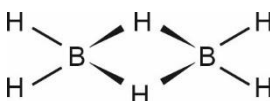
Both B and N are sp^2 hybridised.

94. Answer (1)

Highly poisonous nature of CO arises because of its ability to form a complex with haemoglobin, which is about 300 times more stable than oxygen-haemoglobin complex.

95. Answer (3)

B_2H_6 contains $3\text{C} - 2\text{e}$ bond



96. Answer (1)

$$[\text{H}^+] = \frac{0.2 \times V + 0.1 \times 2 \times V}{V + V} = 0.2 \text{ M}$$

$$\therefore \text{pH} = -\log(0.2) = 0.7$$

97. Answer (2)

Element	Atomic radius (pm)
B	88
Al	143
Ga	135
In	167

98. Answer (2)

$\text{HNO}_3 + \text{HCl}$ will not show common ion effect due to strong acidic nature of HNO_3 and HCl .

99. Answer (4)

$$E_{\text{cell}}^{\circ} = E_{\text{cathode}}^{\circ} - E_{\text{anode}}^{\circ}$$

$$= -0.25 - (-0.76)$$

$$= 0.51 \text{ V}$$

100. Answer (1)

- $\text{Hg}_2(\text{I})\text{Cl}_2$ is the reduced form of $\text{Hg}(\text{II})\text{Cl}_2$.
- Aurous chloride and Auric chloride are written as $\text{Au}(\text{I})\text{Cl}$ and $\text{Au}(\text{III})\text{Cl}_3$ respectively.

BOTANY

SECTION - A

101. Answer (4)

Lysosomes are membrane bound vesicular structure formed by the process of packaging in the golgi apparatus.

102. Answer (2)

In *Amoeba*, the contractile vacuole is important for osmoregulation and excretion.

103. Answer (3)

The M phase usually ends with division of cytoplasm *i.e.*, cytokinesis.

104. Answer (2)

G_1 phase is the interval between mitosis and initiation of DNA replication.

In G_0 phase cell remains metabolically active but do not proliferate unless called on to do so.

In the amount of nucleic acid (DNA) doubles in S-phase.

105. Answer (3)

The important function of Golgi apparatus is to process, package and transport the materials for secretions. It is closely associated with ER in structural as well as functional aspects.

106. Answer (3)

Lysosomes are rich in hydrolytic enzymes that are capable of digesting carbohydrates, proteins, lipids and nucleic acid.

107. Answer (2)

During pachytene, crossing over occurs between non-sister chromatids of the homologous chromosomes which is an enzyme mediated process and the enzyme involved is called recombinase.

108. Answer (2)

Condensation and coiling of chromatin fibres begin during leptotene.

109. Answer (1)

Ribosomes, made up of ribosomal RNAs and proteins, are the cellular components that carryout protein synthesis.

110. Answer (4)

Meiosis increases the genetic variability in the population of organisms from one generation to the next.

111. Answer (3)

Each oxysome is differentiated into base, stalk and headpiece. The headpiece contains ATP synthetase which brings about oxidative phosphorylation coupled with release of ATP.

112. Answer (2)

In S-phase, the amount of DNA doubles but number of chromosomes remain same.

Cell with 20 chromosomes and 32 C DNA in G_1 phase will have 20 chromosomes and 64 C DNA after S-phase.

113. Answer (3)

β -oxidation of fats occurs in mitochondria.

114. Answer (3)

Sister chromatids separate during Anaphase II.

115. Answer (1)

Mitotic divisions in the meristematic tissues - the apical and the lateral meristem, result in a continuous growth of plants throughout their life.

116. Answer (3)

M-phase is the most dramatic period of cell cycle. Interphase corresponds more than 95% duration of cell cycle.

117. Answer (1)

Cells differ greatly in size, shape and activities. For example, Mycoplasmas, the smallest cells, are only $0.3 \mu\text{m}$ in length, while bacteria could be 3 to $5 \mu\text{m}$. Human red blood cells are about $7.0 \mu\text{m}$ in diameter.

118. Answer (1)

Nuclear division and cytokinesis occur twice during the process of meiosis.

119. Answer (2)

Anaphase is the best stage to study shapes of chromosomes.

Metacentric chromosome appears V-shaped during anaphase.

120. Answer (4)

Terminal ends of chromosomes are called telomeres. Centrioles take part in formation of spindle apparatus. Kinetochore are present around centromere and provide a site for attachment to spindle fibres.

121. Answer (1)

Bacterial flagellum is composed of three parts filament, hook and basal body.

122. Answer (4)

Ends of chromosomes are called telomeres. They seal the ends of chromosomes. Chromatids are held together at one point called centromere. On the sides of centromere, disc shaped structures are present known as kinetochores.

In Acrocentric chromosome, one extremely short and one very long arm is present.

123. Answer (2)

Nucleolus is a site for active ribosomal RNA synthesis.

124. Answer (3)

The isolated lysosomal vesicles have been found to be very rich in almost all types of hydrolytic enzymes optimally active at the acidic pH.

125. Answer (4)

In telophase chromosomes cluster at opposite spindle poles and their identity is lost as discrete elements.

126. Answer (2)

Pachytene stage, crossing over leads to recombination of genetic material.

127. Answer (2)

Chiasmata forms in diplotene stage. Cells at the end of prophase, when viewed under the microscope, do not show golgi complexes, ER, nucleolus and the nuclear envelope.

128. Answer (2)

The prophase I is subdivided into five stages which are arranged in following sequence.

Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.

129. Answer (3)

Cell theory was given by Schleiden and Schwann. Golgi apparatus was first observed by Camillo Golgi in 1898. Ribosomes were discovered by George Palade. Rudolf Virchow first explained that cells divided and new cells are formed from pre-existing cells (*Omnis cellula-e-cellula*).

130. Answer (4)

Meiosis increases the genetic variability in the population of organisms from one generation to the next. Variation are very important for the process of evolution.

131. Answer (2)

The complete disintegration of the nuclear envelope marks the start of second phase of mitosis (i.e., metaphase)

132. Answer (1)

Pumps are proteins that use energy to transport substances across the cell membrane.

133. Answer (4)

In oocytes of some vertebrates, diplotene can last for months or years.

134. Answer (2)

Ulothrix, a green alga, is a eukaryotic organism.

135. Answer (1)

Anaphase is the best stage of the cell cycle to study shape of the chromosomes.

SECTION - B

136. Answer (4)

There is no membrane boundary for a centriole, but it is surrounded by pericentriolar satellites, also known as massule (microtubule generator or MTG)

137. Answer (4)

Mitochondria are the sites of aerobic respiration. They produce cellular energy in the form of ATP, hence they are called 'Power houses' of the cell.

138. Answer (2)

Aleuroplasts store proteins while elaioplasts store oils and fats.

139. Answer (2)

In prophase, the DNA strands get untangled to form compact mitotic chromosome.

140. Answer (3)

A tetrad has four chromatids but they are of two homologous chromosomes.

141. Answer (3)

In plant cell middle lamella is chiefly made up of calcium pectate.

142. Answer (1)

In some organisms, karyokinesis is not followed by cytokinesis as a result of which multinucleate condition arises leading to formation of syncytium.

143. Answer (3)

The given figure is of fluid mosaic model of plasma membrane.

A - Sugar

- B - Peripheral protein
C - Phospholipid bilayer
D - Integral protein

144. Answer (3)

Membrane proteins can be classified as integral and peripheral proteins depending on the ease of extraction.

145. Answer (3)

Eukaryotes show intracellular compartmentalisation *i.e.*, membrane bound cell organelles.

146. Answer (2)

The interphase lasts more than 95% of the total duration of cell cycle.

When duration of cell cycle is 120 minutes then its duration of interphase should be more than 95% of 120 minutes, *i.e.*, = more than 114 minutes

147. Answer (2)

During mitosis, number of chromosomes in parent and progeny cells remain the same.

Hence, it is also called equational division.

148. Answer (2)

Since, $n = 4$ (given)

$$\therefore 2n = 8$$

Therefore, the cell has 4 bivalents on equatorial plate during metaphase-I.

Bivalent is the complex formed by a pair of synapsed homologous chromosomes.

149. Answer (4)

- In S-phase, centriole duplicates in cytoplasm of animal cells.
- Centrioles are absent in plant cell.
- DNA replication occurs in S-phase.
- In G_2 phase of cell cycle, semi-autonomous cell organelles duplicates and tubulin proteins are synthesized.

150. Answer (1)

During mitotic anaphase the centromere splits, chromatids separate and move to the opposite poles.

ZOOLOGY

SECTION - A

151. Answer (3)

Humans have two lungs which are covered by double layered pleura, with pleural fluid between them. The outer pleural membrane is in close contact with the thoracic lining whereas the inner pleural membrane is in contact with the lung surface.

Thoracic vertebrae are the structural constituents of the thoracic cage.

152. Answer (4)

Exchange of O_2 and CO_2 at the sites of exchange of gases is dependent on concentration gradient, solubility of the gases and on the thickness of diffusion membrane.

Rate of diffusion of one gas is independent from the rate of diffusion of another gas.

153. Answer (2)

In a healthy adult man, under normal physiological conditions,

TV = approx. 500 mL

IRV = 2500 mL to 3000 mL

ERV = 1000 mL to 1100 mL

RV = 1100 mL to 1200 mL

TV + ERV + IRV is less than TLC and equal to VC.

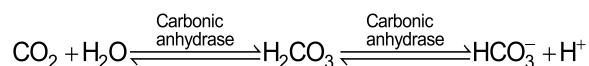
154. Answer (1)

Specialised vascularised structures called gills (branchial respiration) are used by fishes (*e.g.*, *Pterophyllum*), most of the aquatic arthropods (*e.g.*, Prawn) and molluscs (*e.g.*, *Sepia*).

Earthworms use their moist cuticle and insects have a network of tubes (tracheal tubes) for respiration.

155. Answer (4)

RBCs contain a very high concentration of the enzyme, carbonic anhydrase. This enzyme facilitates the following reaction in both the directions.



Zn^{2+} is the co-factor of carbonic anhydrase.

156. Answer (3)

In the alveoli, where there is high pO_2 , low pCO_2 , lesser H^+ concentration and lower temperature, all are the favourable factors for the formation of oxyhaemoglobin.

157. Answer (2)

Pulmonary artery carries deoxygenated blood whereas pulmonary vein carries oxygenated blood.

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O ₂	159	104	40	95	40
CO ₂	0.3	40	45	40	45

158. Answer (4)

Every 100 mL of oxygenated blood can deliver around 5 mL of O₂ to the tissues under normal physiological conditions in an adult human.

Therefore, 300 mL of oxygenated blood will deliver around 15 mL of O₂ to the tissues.

Also, every 100 mL of deoxygenated blood deliver approximately 4 mL of CO₂ to the alveoli.

So, 400 mL of deoxygenated blood will deliver around 16 mL of CO₂ to the alveoli.

159. Answer (4)

During forced expiration, contraction of internal inter-costal muscles occurs leading to the pulling of ribs in downward and inward direction, decreasing the volume of thoracic cavity.

As the volume of thoracic cavity decreases, the pressure inside thoracic cavity increases than the atmospheric pressure, leading to expiration.

160. Answer (1)

The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part of human respiratory system whereas alveoli and their ducts form the exchange part of the respiratory system. The trachea, primary, secondary and tertiary bronchi and initial bronchioles are supported by incomplete cartilaginous rings.

161. Answer (2)

Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the first child.

162. Answer (4)

Simple organisms like sponges (*e.g.*, *Sycon*) and coelenterates (*e.g.*, *Hydra* and *Obelia*) circulate water from their surroundings through their body cavities to facilitate the cells to exchange essential substances. Complex organisms (*e.g.*, *Pheretima*) use special fluids within their bodies for such exchange.

163. Answer (2)

About 70 per cent filling of ventricles takes place during joint diastole. Both the atria undergo a simultaneous contraction during atrial systole. This increases the flow of blood into the ventricles by about 30%.

164. Answer (3)

Plasma is a straw coloured, viscous fluid constituting nearly 55 per cent of the blood and formed elements constitute about 45 per cent of the blood.

165. Answer (1)

Clot/coagulum is formed by a network of threads called fibrins which are derived from inactive fibrinogens.

166. Answer (4)

In fishes, the heart pumps out deoxygenated blood which is oxygenated by the gills and supplied to the body parts from where deoxygenated blood is returned to the heart (single circulation).

167. Answer (1)

The nodal musculature has the ability to generate action potentials without any external stimuli, *i.e.*, it is autoexcitable.

However, the number of action potentials that could be generated in a mixture vary at different parts of the nodal system. The SAN can generate the maximum number of action potentials *i.e.*, 70-75 min⁻¹ and is responsible for initiating and maintaining the rhythmic contractile activity of the heart.

168. Answer (3)

As the blood passes through the capillaries in tissues, some water along with many water soluble substances move out into the spaces between the cells of tissues leaving the larger proteins and most of the formed elements in the blood vessels. This released out fluid is called lymph.

169. Answer (3)

Basophils are the cells responsible for secretion of histamine and serotonin.

Monocytes and lymphocytes are the agranulocytes.

170. Answer (4)

Fishes have a 2-chambered heart with an atrium and a ventricle. In fishes, blood get oxygenated by gills and heart pumps out deoxygenated blood.

In humans, oxygenated and deoxygenated blood received by the left and right atrium respectively, passes into the ventricles of the same side which pump it out without any mixing up.

171. Answer (4)

Silicosis is common occupational respiratory disorder. In asthma, there is difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

172. Answer (2)

A sigmoid curve is obtained when percentage saturation of haemoglobin with O_2 is plotted against the pO_2 .

This sigmoid curve is known as oxygen dissociation curve.

173. Answer (2)

The epithelial tissue present in the alveoli of human lungs is simple squamous epithelium.

If the cuboidal or columnar cells bear cilia on their free surface they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles.

174. Answer (2)

Neural signals through the sympathetic nerves can increase the rate of heart beat, the strength of ventricular contraction and thereby the cardiac output.

Also, adrenal medullary hormones like epinephrine and norepinephrine can increase the cardiac output.

175. Answer (3)

A bicuspid/Mitral valve guards the opening between the left atrium and the left ventricle. Mitral valve will evert out into atrium during ventricular systole in this condition.

176. Answer (2)

Blood vessels in between left ventricle and right atrium constitute systemic circulation through which blood flows.

177. Answer (1)

A bundle of nodal fibres, AV bundle continues from the AVN which passes through the atrio-ventricular septa to emerge on the top of the interventricular septum and immediately divides into a right and left bundle, which transmit action potential to the entire ventricular musculature through purkinje fibers.

178. Answer (2)

Upon holding the breath, the amount of CO_2 in the body increases, which leads to a change in the pH of the blood.

This changing pH gets recognised by the chemoreceptors present in blood vessels activates respiratory rhythm centre in brain, signals the muscles associated with respiratory system and leads to an urge to breath.

179. Answer (4)

Tunica intima is the inner layer of blood vessels made up of squamous endothelium. The smallest vessels called capillaries have walls which are one-celled thick.

180. Answer (1)

A specialised nodal tissue embedded in the lower corner of the right atrium, close to atrio-ventricular septum, delays the spreading of impulses to heart apex for about 0.1 seconds. This delay allows the atria to empty completely.

181. Answer (3)

Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

It is not same as cardiac arrest (when heart stops beating) or a heart attack (when the heart muscle is suddenly damaged by an inadequate blood supply).

182. Answer (3)

An injury or trauma stimulates the platelets in blood to release certain factors which activate the mechanism of coagulation. Platelets, also known as thrombocytes are 1.5 lakh - 3.5 lakh per mm^3 of blood.

183. Answer (4)

Factors for clotting of blood are present in the plasma in an inactive form. Plasma without the clotting factors is called serum.

184. Answer (1)

In a standard ECG, the P-wave represents the electrical excitation (or depolarisation of the atria), the QRS complex represents the depolarisation of the ventricles. The T-wave represents the repolarisation of ventricles and end of T-wave marks the end of the ventricular systole.

185. Answer (4)

In an open type of blood circulation, the blood is pumped out of the heart and the cells and tissues are directly bathed in it.

Hemichordates, arthropods, non-cephalopod molluscs have an open type of blood circulation whereas most of annelids and chordates have closed type of blood circulation.

SECTION - B

186. Answer (3)

A special neural centre in the medulla oblongata can moderate the cardiac function through autonomic nervous system (ANS).

187. Answer (1)

Cardiac output = $SV \times HR$

$$\begin{aligned}\text{Stroke volume} &= \text{EDV} - \text{ESV} \\ &= 150 \text{ mL} - 100 \text{ mL} \\ &= 50 \text{ mL}\end{aligned}$$

$$\text{HR} = \frac{\text{CO}}{\text{SV}} = \frac{5000 \text{ mL/minute}}{50 \text{ mL}}$$

$$= 100 \text{ beats per minute}$$

188. Answer (3)

In well-trained athletes, stroke volume can also increase significantly allowing them to achieve cardiac output during strenuous exercise upto six-seven times greater than their resting values.

189. Answer (4)

Each artery and vein consists of three layers: an inner lining of squamous endothelium, the tunica intima, a middle layer of smooth muscle and elastic fibres, the tunica media, and an external layer of fibrous connective tissue with collagen fibres, the tunica externa.

190. Answer (2)

Rh antibodies from the mother (Rh-ve) can leak into the blood of the foetus (Rh +ve) and destroy the foetal RBCs.

191. Answer (1)

- A – Squamous epithelium of alveolar wall
- B – Basement substance
- C – Red blood cell
- D – Endothelium of blood capillary

192. Answer (2)

At high altitude, atmospheric pO_2 level is less and hence, more RBCs are needed to transport the required amount of O_2 to the tissues to survive.

193. Answer (4)

Intrapleural fluid is present between the two pleural layers of lungs. This fluid helps in maintaining the negative pressure in the pleural cavity as during inhalation the lungs expands and does not collapse because of presence of this fluid.

194. Answer (4)

Residual volume is the volume of air remaining in the lungs even after a forcible expiration. This averages 1100 mL to 1200 mL.

195. Answer (3)

Blood is the medium of transport for O_2 and CO_2 . About 97% of O_2 is transported by RBCs in the blood. The remaining 3% of O_2 is carried in a dissolved state through the plasma. Nearly 20-25% of CO_2 is transported by RBCs whereas 7% of it is carried in a dissolved state through plasma.

196. Answer (3)

A chemosensitive area, situated adjacent to the respiratory rhythm centre is highly sensitive to CO_2 and hydrogen ions.

Increase in CO_2 and hydrogen ions can activate this chemosensitive area. The respiratory rhythm centre is located in the medulla oblongata of the brain.

197. Answer (4)

We know, Cardiac output = Heart rate $\times$ Stroke volume

Given, 7000 mL = Heart rate $\times$ 70 mL

$$\Rightarrow \text{Heart rate} = \frac{7000}{70} = 100$$

$$\text{Now, duration of cardiac cycle} = \frac{60}{\text{Heart rate}} = \frac{60}{100}$$

$$= 0.6 \text{ seconds}$$

198. Answer (3)

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

199. Answer (2)

During atrial systole, both bicuspid/Mitral and tricuspid valves are open but during ventricular contraction, they get close.

Also, during ventricular diastole AV valves open. Semilunar valves open when ventricles undergo contraction and are located between right ventricle and pulmonary artery and between left ventricle and aorta respectively.

200. Answer (4)

A unique vascular connection exists between the digestive tract and liver called hepatic portal system. The hepatic portal vein carries blood from intestine to the liver before it is delivered to the systemic circulation. It ensures that the food ingested is processed by the liver first before entering the systemic circulation. This way, ingested toxins are detoxified by hepatocytes.